installation of rigid metal conduit, fittings, mounting hardware, ground wire, ground clamp, insulating clevis, ground plate and branch circuit wiring to the handhole at the base of the pole; supply and installation of a working pad including additional fill as required (see SS Drawing SP635-2.4.20); and all other labour, equipment and materials necessary to complete the installation.

635.21 Wiring – All wiring shall be installed in accordance with the Drawings or as directed by the Ministry Representative.

All conductors shall be stranded copper, RW90 XLPE insulated unless otherwise noted.

Conductor gauges (AWG) shall be as specified on the Drawings.

Shielded detector loop and telephone demarcation cables shall be 2 conductor No. 18 Ministry pre-approved product. Shielded detector loop cables shall run continuous with no splices, from the traffic controller (or other type of control equipment) to the respective detector loop conductor.

Single conductor colour coding and labelling is required and shall be as noted on SS Drawings SP635-2.5.7 and SP635-2.5.8. Conductor colour coding shall also meet the requirements of the Canadian Electrical Code.

Where specifically noted on the Drawings signal cable shall be used. Signal cable colour coding and conductor designations shall be as noted on SS Drawing SP635-2.5.9. Where signal cable is used all conductor splices shall be made in pole hand holes in accordance with SS Drawing SP635-2.5.10.

Each traffic and pedestrian signal head section and luminaire shall be wired separately to the base of the pole. A separate neutral and bond conductor shall run to each signal head and luminaire from the base of the pole. The neutral and bond conductors shall be bundled with the feeder circuits.

Luminaire conductor wiring and fusing in pole handholes shall conform to SS Drawing SP635-2.5.6.

Wiring inside junction boxes and vaults shall conform to SS Drawings SP635-2.5.1 through SP635-2.5.5.

Conductor splices shall be secured with solderless "Marrette" type connectors. Where the number and/or size of conductors exceed the capacity of the Marrette, split bolt connectors shall be used.

All wiring shall be neatly bundled and labelled in all junction boxes, vaults, traffic controllers, handholes at pole bases, and service panels.

Sealing of connections shall be performed using one of the following methods:

(a) Double dipping using 3M "Scotchcote" or approved alternate. Dipping shall be performed strictly adhering to the manufacturer's specifications.

(b) Each conductor shall have a wrap of the self-holding tape (3M Linerless Rubber Splicing Tape, Steadfast 8 Ounce Splicing tape or approved alternate), then the complete splice shall be wrapped. PVC tape shall then be applied to cover the complete splice.

If conductor connections require the use of split bolts or similar style devices due to wire size, then the splice shall be completely covered with Duct Seal to form a ball over the connection. This Duct Seal shall be thick enough to prevent sharp ends of the conductors and/or points of the connector from protruding through the taped connection. Once the duct seal has been applied, the splice shall be taped with self-holding and PVC tape as described in SS 635.21(b).

All luminaires and signal heads shall be bonded with a No. 12 RW90 conductor. Steel junction box lids and steel vault lids shall be bonded in accordance with SS Drawings SP635-2.5.1 through SP635-2.5.5.

635.21.02 Payment – Payment for the installation of WIRING will be at the Lump Sum Price for all underground wiring included in the Work.

The Lump Sum Price shall include all costs for the supply and installation of wiring and cables located in the underground conduit system (unless otherwise noted) and splices; labelling of underground conductors; and all other labour, equipment and materials necessary to complete the installation.

635.22 Traffic Counter Stations – Traffic counter stations shall be short-duration type or permanent type.

Short duration traffic counter stations shall be constructed as noted on the Drawings and shall be installed in accordance with SS Drawing SP635-2.6.1.

Perforated square steel tubing shall conform to SS 635.30.

Permanent traffic counter stations shall be constructed as noted on the Drawings and shall be installed in accordance with SS Drawing SP635-2.6.2 and SP636-2.6.3.

The Contractor will install all pole mount cabinets. The Ministry Electrical Maintenance Contractor will connect all field wiring in cabinets, trouble-shoot problems in the cabinet and undertake all equipment set-up.

635.22.01 Payment – Payment for the installation of TRAFFIC COUNTER STATIONS will be at the Unit Price per traffic counter station.

The Unit Price shall include all costs for the supply and installation of perforated steel tubing, above ground conduit and junction boxes, connectors; supply and installation of rigid metal conduits, fittings, mounting hardware and branch circuit wiring to the handhole at the base of pole for pole mount cabinets; installation of counter cabinet including the pole where a separate pole is required; Supply and installation of W-55 signs and concrete pad in front of

the controller door; and all other labour, equipment and materials necessary to complete the installation.

635.23 Controllers – Type B, C, M, S and P6 traffic controller cabinets shall be installed in accordance with SS Drawings SP635-2.7.1 through SP635-2.7.4.

The Ministry Electrical Maintenance Contractor will install all base mount traffic controllers and cabinets. The Contractor will install all pole mount cabinets. The Ministry Electrical Maintenance Contractor will connect all field wiring in controller cabinets, trouble-shoot problems in the cabinet, enter all signal timings, undertake modifications to existing signal phasing and timings, start-up controller and complete Ministry signal turn-on sheets.

The Contractor shall be on hand to assist with controller start-up and make any repairs to field wiring or hardware as required. The Contractor shall provide traffic control and assistance during the controller start-up.

The Contractor shall coil and label 2 m of each conductor inside the traffic controller for connections by the Ministry Electrical Maintenance Contractor. The Contractor shall verify that all traffic and pedestrian signal phases are wired as shown on the Drawings and that all circuits are tested prior to activation. The Contractor shall verify that all signal and pedestrian phases are properly colour coded and labelled with identification tags.

635.23.01 Payment – Payment for the installation of each CONTROLLER will be at the Lump Sum Price for that controller.

The Lump Sum Price shall include all costs for the installation of pole mount cabinets and installation of the pole where a separate pole is required, rigid metal conduits, fittings, mounting hardware and branch circuit wiring to the handhole at the base of pole for pole mount cabinets; testing of all circuits; labelling of all conductors; traffic control and all other labour, equipment and materials necessary to complete the installation.

635.24 Detector Loops – Detector loops shall be constructed as noted on the Drawings and shall be installed in accordance with SS Drawings SP635-2.8.1 through SP635-2.8.17. Loop check sheets shall be completed and submitted to the Ministry Representative prior to signal start-up.

635.24.01 Payment – Payment for DETECTOR LOOPS will be at the Unit Price per loop.

The Unit Price shall include all costs for layout of the loop, asphalt cutting and preparation of pavement cuts; supply and installation of loop conductors or preformed loops as applicable, to the underground junction box, traffic counter post or cabinet; supply and installation of backer rod, sand and loop sealant; and all other labour, equipment and materials necessary to complete the installation.

635.25 Repairing Galvanized Surfaces – Any spots where the galvanized finish is damaged due to drilling, tapping, reaming or welding and any surface damage incurred during transportation and erection shall be refinished with cold galvanizing compound. Cold galvanizing compound shall be selected from the Ministry Recognized Products List under the category 'Additional Paint Coatings – Zinc Rich Touch-up Paints and Primers'. The application of cold galvanizing compound shall conform to the Manufacturer's instructions, 'Notes for Use' on the Recognized Products List, and the following:

(a) The surface shall be mechanically cleaned with a wire brush or grinder and chemically cleaned to remove all welding flux, paint, grease, oil, rust, scale or other detrimental foreign matter.

(b) The surface shall be absolutely dry and the ambient temperature shall be over 10°C.

(c) Uniform coats shall be applied. Each coat shall be as thick as possible without causing runs on the finished surface.

All costs for the repair of galvanized surfaces will be considered incidental to the Work.

635.26 Flasher Luminaires – Flasher luminaires shall be mounted on perforated square steel tubing in accordance with SS Drawings SP635-2.9.1 through SP635-2.9.3.

Flasher luminaires shall be mounted on poles in accordance with SS Drawings SP635-2.9.4 through SP635-2.9.6.

All perforated square steel tubing shall be in accordance with SS 635.30.

635.26.01 Payment – Payment for the installation of Flasher Luminaires on Perforated Square Steel Tubing will be at the Unit Price for a one or two sign unit.

The Unit Price shall include all costs for the supply and installation of perforated square steel tubing, mounting hardware, wiring to the junction box nearest to the flasher post, ty-raps and connectors, signs, flasher luminaires, lamps; and all other labour, equipment and materials necessary to complete the installation.

635.26.02 Payment – Payment for the installation of Flasher Luminaires on Poles will be at the Unit Price for each item.

The Unit Price shall include all costs for the supply and installation of all fittings, hardware and wiring to the underground junction box nearest the pole, signs, flasher luminaires and lamps; and all other labour, equipment and materials necessary to complete the installation.

PART D – SIGNING

Note: Removal of existing signs shall be in accordance with SS 200.

635.27 Overhead Signs – Overhead signs are categorized in two classes:

- small overhead signs are 1200 mm x 900 mm or smaller and made from sheet aluminum;
- large overhead signs are larger than 1200 mm x 900 mm and made from sheet aluminum or extruded aluminum.

Note: Signs are no longer made from plywood.

Small overhead signs are generally mounted on signal poles and shall be installed in accordance with SS Drawings SP635-3.2.2 through SP635-3.2.6.

Large overhead sheet aluminum signs are installed on sign poles in accordance with SS Drawings SP635-3.3.1 through 3.3.4, unless otherwise noted. Large overhead extruded aluminum signs are installed on sign poles in accordance with SS Drawings SP635-3.3.5 through SP635-3.3.15, and SP535-3.3.18 unless otherwise noted. Sign poles shall be installed in accordance with SS Drawings SP635-3.1.1 through SP635-3.1.22.

Sign lighting is used only when specified by the Ministry. If specified, it shall be installed in accordance with SS Drawing SP635-3.3.2 and SP635-3.3.16, and SS 635.19.03.

Overhead signs shall be securely installed on sign pole structures.

All horizontal supports and signs shall be level and vertical supports plumb.

Sign Poles shall be installed in accordance with SS 635.19.

Advance warning signs shall be extruded aluminum and shall be installed in accordance with SS Drawings SP635-3.3.5 through SP635-3.3.8.

All signs shall be installed in accordance with SS 635.34.

635.27.01 Payment – Payment for the installation of each OVERHEAD SIGN POLE will be at the Lump Sum Price for that sign pole.

The Lump Sum Price shall include all costs for installation of sign poles and arms, overhead signs complete with T Section mounting brackets and clips, advance warning flasher equipment, sign luminaries and mounting hardware; small overhead signs and mounting brackets, wiring to the handhole at the base of the pole, fuse holders, fuses, junction boxes on pole arms, splices; and all other labour, equipment and materials necessary to complete the installation.

Where small overhead signs are installed on signal poles, payment will be made under SS 635.19.09.

635.28 Breakaway Sign Structures – Breakaway sign structures shall be installed in accordance with SS Drawings SP635-3.4.1 through SP635-3.4.10.

After concrete bases are installed, the Contractor shall survey the finished base elevations to determine the correct leg lengths. The fabricator shall cut legs to the correct lengths.

Legs and columns are to be transported in an unstressed manner.

Breakaway structures shall be installed with the legs and columns plumb, and the battens and signs level.

The Contractor shall tighten all bolts and nuts with a torque wrench to the torque specified on the Drawings.

Signs shall be sheet aluminum or extruded aluminum, as specified.

Extruded aluminum signs shall be installed in accordance with SS Drawings SP635-3.4.7 through SP635-3.4.10 and SS 635.34.

Extruded aluminum signs on breakaway sign structures shall not be illuminated.

All sign installations shall be in accordance with SS 635.34.

635.28.01 Payment – Payment for the installation of each BREAKAWAY SIGN STRUCTURE will be at the Lump Sum Price for that breakaway sign structure.

The Lump Sum Price shall include all costs for the installation of breakaway sign legs, columns, stub posts, fuse and connection joints, sign or signs, sign luminaires and mounting brackets, battens, painting (where required), sign mounting bolts and hardware for signs; and all other labour, equipment and materials necessary to complete the installation.

635.29 Wood Post Sign Structures – Wood post sign structures shall be installed in accordance with SS Drawings SP635-3.5.1 through SP635-3.5.9.

Excavations for wood posts shall be wide enough to allow for proper compaction of backfill around the wood posts. Wood posts shall be embedded in the ground to the depths indicated on SS Drawings SP635-3.5.1 through SP635-3.5.3.

Wood post excavations shall be backfilled using the excavated material except where excavated material is ruled unacceptable by the Ministry Representative. Where new material is required, 25 mm Well Graded Base Course Aggregate material shall be used as specified in SS 635.16. Unacceptable materials shall be disposed of as specified in SS 635.17.

Wood posts shall be Douglas Fir/Larch, No. 1 Grade or pressure treated, surfaced four sides, and shall be supplied in complete lengths without splices. Posts shall be straight and free of cracks.

All wood products used in wood sign post structures shall be preservative treated and meet the requirements of SS 908.

Wood posts shall be installed plumb and at the proper offset and elevation.

Backfill material shall be placed in layers not exceeding 150 mm compacted thickness and shall be compacted to a minimum 100% of the Standard Proctor maximum dry density obtained by ASTM D698. Layer thickness and moisture content of the material shall be adjusted as necessary to achieve compaction.

All areas excavated and backfilled shall be restored to their original condition.

Signs on single wood post structures shall be sheet aluminum and shall be installed in accordance with SS Drawing SP635-3.5.1 and SS 635.34.

Signs on multiple wood post structures shall be sheet aluminum or extruded aluminum.

Extruded aluminum signs shall be installed in accordance with SS Drawings SP635-3.5.6 through SP635-3.5.9 and SS 635.34.

All areas around the post shall be fully restored to their original condition.

635.29.01 Payment – Payment for the installation of SINGLE WOOD POST SIGN STRUCTURES will be at the Unit Price for each wood post sign structure.

The Unit Price shall include all costs of Type D excavation, other than concrete or asphalt removal; supply and installation of wood posts, painting (where required), sign mounting bolts and hardware and signs; placing and compaction of excavated materials as backfill; removal of excess excavated material; restoration; and all other labour, equipment and materials necessary to complete the installation.

Payment for excavation of Type A material (solid rock) and de-watering of excavations will be made in accordance with GC 38.00 Changes to Work.

Where excavated material is ruled unacceptable for backfill, payment for 25 mm Well Graded Base Course Aggregate will be made under SS 635.16.01.

635.29.02 Payment – Payment for the installation of each MULTI POST WOOD POST SIGN STRUCTURE will be at the Lump Sum Price for that wood post sign structure.

The Lump Sum Price shall include all costs of Type D excavation, other than concrete or asphalt removal; supply and installation of wood posts, painting (where required), sign mounting bolts and hardware, battens for signs, sign(s) and aluminum angle sign supports for extruded aluminum signs; placing and compaction of excavated materials as backfill; removal of excess excavated material; restoration and all other labour, equipment and materials necessary to complete the installation.

Payment for excavation of Type A material (solid rock) and de-watering of excavations will be made in accordance with GC 38.00 Changes to Work.

Where excavated material is ruled unacceptable for backfill, payment for 25 mm Well Graded Base Course Aggregate will be made under SS 635.16.01.

635.30 Perforated Square Steel Sign Post Structures – Perforated Square Steel Sign Post shall be supplied in accordance with SS Drawings SP635-3.6.1 through SP635-3.6.4.

Sign posts shall be perforated square steel tubing. The perforated square steel tubing will be required in different outside dimensions as noted on the Drawings. The perforated square steel tubing shall be formed from 12 gauge hot rolled steel, conforming to ASTM A1011 Grade 50 The tubing shall be hot dipped galvanized conforming to ASTM A653 Designation G-90 or CSA G164. Steel tubing shall have 7/16" holes on all four sides at 1" centres.

Galvanized perforated square steel tubing shall be supplied in continuous lengths, with no splices, and shall be field cut to suit the installation. All field cuts shall be painted with cold galvanizing compound in accordance with SS 635.25.

Perforated square steel tubing sign posts shall be installed on concrete bases or direct buried in accordance with SS Drawing SP635-3.6.1.

Perforated square steel tubing posts shall be installed plumb.

Signs installed on single or double perforated square steel sign posts shall be sheet aluminum. Sheet aluminum signs shall be installed in accordance with SS Drawings SP635-3.6.1 through SP635-3.6.4 and SS 635.34.

635.30.01 Payment – Payment for the installation of SINGLE PERFORATED SQUARE STEEL SIGN POST STRUCTURES will be at the Unit Price for each structure.

The Unit Price shall include all costs for the supply and installation of perforated square steel tubing, mounting hardware, sign(s); and all other labour, equipment and materials necessary to complete the installation.

635.30.02 Payment – Payment for DOUBLE PERFORATED SQUARE STEEL SIGN POST STRUCTURES will be at the Unit Price for each sign structure.

The Unit Price shall include all costs for the supply and installation of perforated square steel tubing; mounting hardware, sign(s); and all other labour, equipment and materials necessary to complete the installation.

635.31 Round Steel Sign Post Structures – Round Steel sign posts shall be installed in accordance with SS Drawings SP635-3.8.1 through SP635-3.8.3.

Barrier sign posts shall be mounted on barrier stands, which shall be securely bolted to concrete roadside or concrete median barriers.

Posts and pipe sleeves shall be round Schedule 40 steel pipe conforming to ASTM A53, Grade A or B, Type E or S. The pipe shall be hot dip galvanized in accordance with CSA G164. Pipe shall be field cut to lengths to suit sign mounting heights as noted on the Drawings or as directed by the Ministry Representative. All field cuts in galvanized steel pipes shall be coated with cold galvanizing compound in accordance with SS 635.25.

Signs shall be installed in accordance with SS 635.34.

635.31.01 Payment – Payment for the installation of ROUND STEEL SIGN POST STRUCTURES will be at the Unit Price for each sign post.

The Unit Price shall include all costs for the supply and installation of sign or signs and barrier stands, sign posts and mounting hardware, breakaway devices, where warranted, pipe sleeves including trenching and backfilling where posts are installed in concrete sidewalks; and all other labour, equipment and materials necessary to complete the installation.

635.32 Sign Mounting on the Side of Poles – Signs mounted on the side of steel poles shall be installed in accordance with SS Drawing SP635-3.9.1.

Signs shall be securely attached to poles.

Signs shall be installed in accordance with SS 635.34.

Holes drilled in galvanized steel poles shall be coated with cold galvanizing compound in accordance with SS 635.25.

635.32.01 Payment – Payment for the installation of SIGN INSTALLATION ON THE SIDE OF POLES will be at the Unit Price for signs installed on each pole.

The Unit Price shall include all costs for the supply and installation of signs, sign mounting hardware; and all other labour, equipment and materials necessary to complete the installation.

Where multiple signs are to be installed on a pole only one unit item will be paid.

635.33 Delineators – Delineators shall be wood, perforated steel tubing or plastic as noted on the Drawings or as directed by the Ministry Representative.

Wood delineator posts shall be installed in accordance with Drawing SP635-3.10.1.

Perforated square steel tubing delineator posts shall be installed in accordance with SS Drawing SP635-3.10.2.

Where delineators are mounted on a pole, sign post or structure, they shall be installed in accordance with SS Drawing SP635-3.10.3.

Where plastic delineators are specified, they shall be a Ministry preapproved type and shall be installed in accordance with the manufacturer's instructions and SS Drawing SP635-3.10.4.

Delineator post excavation, backfill, type of wood, painting (where required) and installation methods shall be in accordance with SS 635.29.

Perforated square steel tubing shall be in accordance with SS 635.30.

All holes drilled in galvanized surfaces shall be coated with cold galvanizing compound in accordance with SS 635.25.

635.33.01 Payment – Payment for the installation of DELINEATORS will be at the Unit Price for each delineator.

The Unit Price shall include all costs of excavation, other than concrete or asphalt removal; supply and installation of delineator posts and mounting hardware, W-055 background plate and reflectors for perforated sign posts and W-055 reflectors for wood sign posts; placing and compaction of excavated materials as backfill; removal of excess excavated materials; and all other labour, equipment and materials necessary to complete the installation.

635.34 Signs – Signs smaller than 1200 mm x 900 mm shall be sheet aluminum. Signs 1200 mm x 900 mm or larger shall be extruded aluminum.

All sign installations shall meet the requirements of the most current edition of the Ministry *Manual of Standard Traffic Signs and Pavement Markings* available at:

> https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines/traffic-engineering-safety/traffic-signs-markings

The Contractor shall have a current copy of this document on site when installing signs.

Signs shall be completely covered with a suitable grade of polyethylene sheeting from the time they are installed until the roadway is in full operation, unless otherwise directed by the Ministry Representative. The polyethylene sheeting material shall prevent sign messages from being visible.

Signs shall be handled so as not to damage them in any way. Slip sheets between signs shall be removed carefully to avoid damage due to adhesion of the slip sheet to the sign. Signs shall be stored on end on dunnage or racks in a dry, covered area, protected from damage. Damaged signs shall be replaced by the Contractor at the Contractor's expense. Taping, screwing, nailing, gluing, bolting or stapling to sign faces or back is prohibited unless otherwise noted.

The Contractor shall confirm that signs have the correct messaging. The Contractor shall verify that signs are free of cracks, dents or warpage prior to installation. Any flaws

with the sign shall be immediately reported to the Ministry Representative prior to installation.

Signs shall be bolted to the mounting hardware or structure as indicated on the applicable Drawings. The sign panels shall be tightened to eliminate sign movement but not over tightened so as to recess the bolt heads into the sign face.

Sheet aluminum signs larger than 1200 mm x 3000 mm will consist of multiple panels. Each panel shall be correctly aligned so that no gaps exist between sign panels. The bottom of the sign panel shall be installed level.

Sheet aluminum exit number tabs shall attach to extruded aluminum signs in accordance with SS Drawings SP635-3.7.1.

Extruded aluminum signs will be supplied unassembled. Aluminum sign sections shall be assembled and correctly aligned to the satisfaction of the Ministry Representative.

Extruded aluminum signs are supplied in typical widths shown on SS Drawing SP635-3.3.11.

Extruded aluminum exit number tabs shall attach to extruded aluminum signs in accordance with SS Drawings SP635-3.7.2 and SP635-3.7.3.

Patching material shall meet current Ministry Specifications.

Custom sheet aluminum and extruded aluminum signs will be identified with their sign numbers labelled on the back.

The Contractor shall maintain all signs installed as part of the work within the project area for the duration of the Work. Maintenance shall be performed for the duration of the work in accordance with the following:

(a) Sign maintenance shall include the straightening, replacing, repairing, and cleaning of all signs installed as part of the Work. Sign maintenance shall be at the discretion of the Ministry Representative.

(b) The Contractor shall regularly inspect the signs to ensure consistent maintenance for maximum visibility.

(c) Where directed by the Ministry Representative the Contractor shall clean the signs by power washing or steam cleaning using pressures not exceeding 7 MPa and temperatures not exceeding 65°C. Chemical washing of sign faces is not permitted.

All costs for installation and maintenance of signs shall be included in the prices for the applicable items.

CONCRETE BASES AND RELATED DETAILS			
BASE TYPE	SP635 DRAWING	SUPPORTED STRUCTURE	STYLE
A	1.1.2	TYPE 4 POLES	SONOTUBE
B	1.1.2	TYPE 4A AND 5 POLES	SONOTUBE
C/CM	1.1.3/1.1.3a	7.5m, 9.0m AND 11.0m TYPE 2 POLES, TYPES 4, 4A AND 5 POLES	TRAPEZOIDAL
D1	1.1.4 TO 1.1.6	13.5m TYPE 2 POLES	SPREAD FOOTING
D2	1.1.7 & 1.1.8		TRAPEZOIDAL
D3	1.1.9 & 1.1.10		RECTANGULAR
E1	1.1.4 TO 1.1.6	TYPE 1 & 3 POLES	SPREAD FOOTING
E2	1.1.7 & 1.1.8		TRAPEZOIDAL
E3	1.1.9 & 1.1.10		RECTANGULAR
F1	1.1.11 TO 1.1.13	TYPE 6 & 7 POLES	SPREAD FOOTING
F2	1.1.14 & 1.1.15		TRAPEZOIDAL
F3	1.1.16 & 1.1.17		RECTANGULAR
S1	1.1.11 TO 1.1.13	TYPE S POLES	SPREAD FOOTING
S2	1.1.14 & 1.1.15		TRAPEZOIDAL
S3	1.1.16 & 1.1.17		RECTANGULAR
L1	1.1.11 TO 1.1.13	TYPE L POLES	SPREAD FOOTING
L2	1.1.14 & 1.1.15		TRAPEZOIDAL
L3	1.1.16 & 1.1.17		RECTANGULAR
–	1.1.18	ANCHOR BOLT CAGE FOR TYPE 6, 7, & S POLES	–
–	1.1.19	ANCHOR BOLT CAGE FOR TYPE L POLES	–
M1	1.1.20 TO 1.1.22	TYPE M POLES	SPREAD FOOTING
M2	1.1.23 & 1.1.24		RECTANGULAR
–	1.1.25	ANCHOR BOLT CAGE FOR TYPE M POLES	–
H1	1.1.26 TO 1.1.28	TYPE H POLES	SPREAD FOOTING
H2	1.1.29 & 1.1.30		RECTANGULAR
–	1.1.31	ANCHOR BOLT CAGE FOR TYPE H POLES	–
–	1.1.32 TO 1.1.34	BREAKAWAY SIGN STRUCTURES	SPREAD FOOTING
–	1.1.35 & 1.1.36	SIGN POSTS BASES	SONOTUBE
–	1.1.37	POST MOUNTED FLASHER CONCRETE BASE	SONOTUBE
–	1.1.38 TO 1.1.42	CONTROLLER BASE	TRAPEZOIDAL
–	1.1.43 & 1.1.44	ANCHOR BOLT REPAIR PROCEDURES	–
–	1.1.45 TO 1.1.47	STEEL SIGN/FLASHER POST	TRAPEZOIDAL

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D	GENERAL REVISIONS	APR 17
C	TYPE S BASES ADDED	OCT 03
B	DRAWING LIST REVISED	AUG 95
A	GENERAL REVISIONS	AUG 94

CONCRETE BASE INDEX

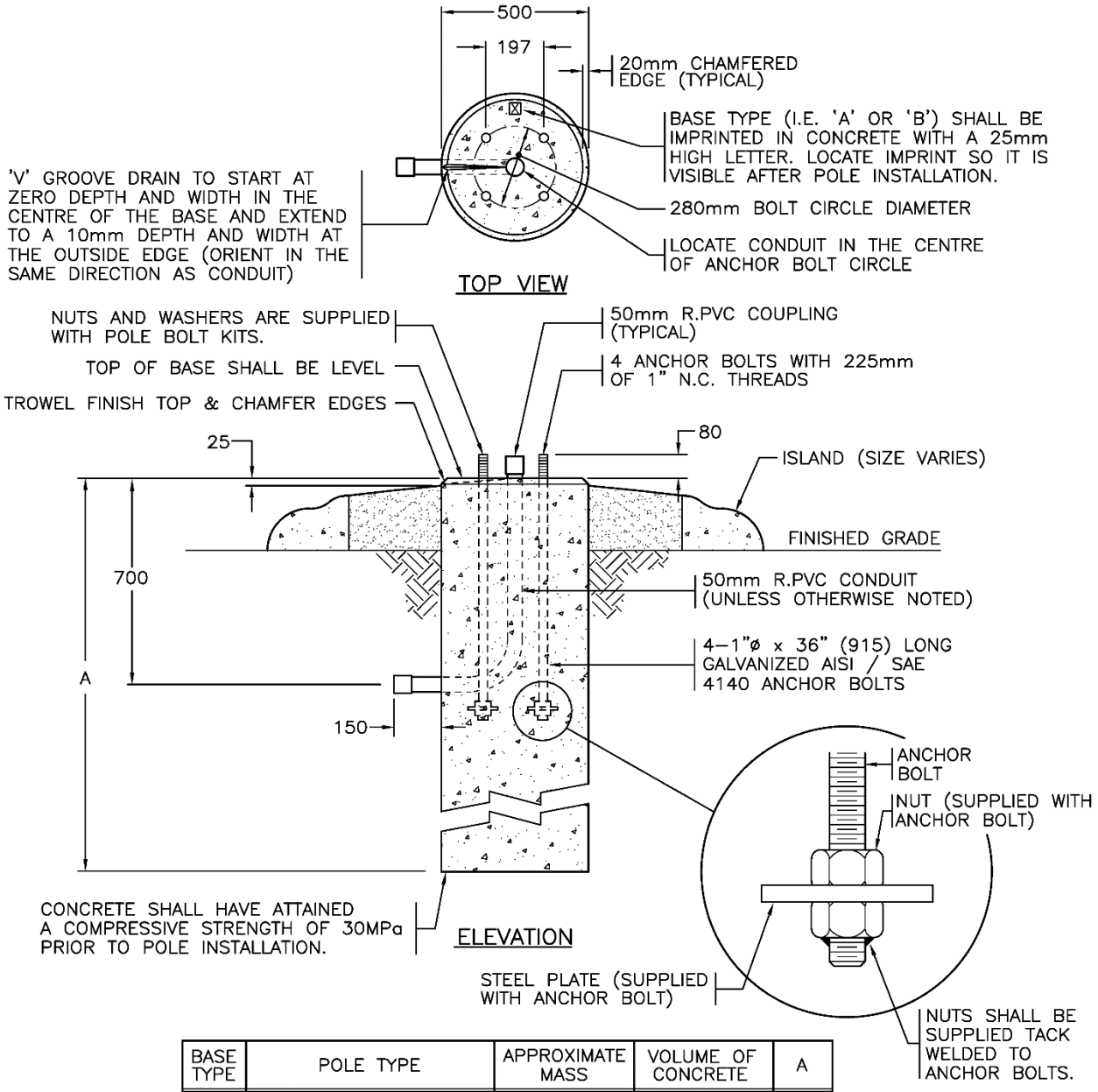
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–1.1.1



BASE TYPE	POLE TYPE	APPROXIMATE MASS	VOLUME OF CONCRETE	A
A	TYPE 4 POLE	620 kg	0.24 m^3	1200
B	TYPE 4A AND 5 POLES	760 kg	0.30 m^3	1500

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. SEE DRAWINGS SP635–1.4.1 TO 1.4.4 FOR BACKFILL REQUIREMENTS.

NOT TO SCALE

BRITISH COLUMBIA Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D	TITLE UPDATED	APR 20
C	GENERAL REVISIONS	APR 17
B	NOTE 4 ADDED	AUG 95
A	GENERAL REVISIONS	AUG 94

TYPES A AND B CONCRETE BASES (PRECAST OR CAST IN PLACE)

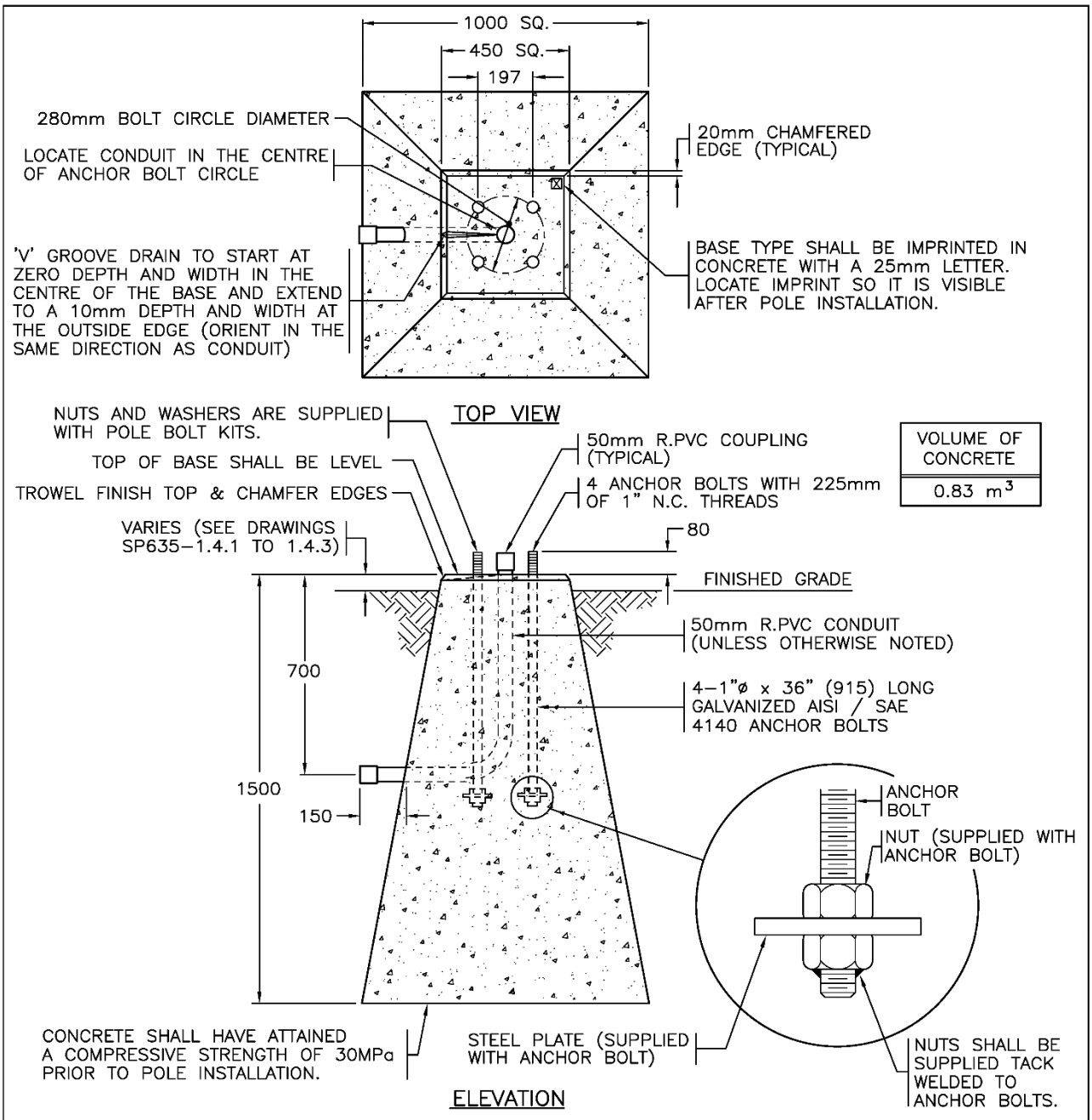
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–1.1.2



VOLUME OF CONCRETE
0.83 m^3

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. SEE DRAWINGS SP635-1.4.1 TO 1.4.4 FOR BACKFILL REQUIREMENTS.

NOT TO SCALE

BASE TYPE	POLE TYPE	APPROXIMATE MASS
C	7.5m, 9.0m AND 11.0m TYPE 2 AND TYPES 4, 4A AND 5 POLES	2000 kg

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D	TITLE UPDATED	APR 20
C	GENERAL REVISIONS	APR 17
B	ELEVATION OF BASE TO FINISHED GRADE REVISED	AUG 95
A	GENERAL REVISIONS	AUG 94

TYPE C CONCRETE BASE (PRECAST OR CAST IN PLACE)

Date Approved

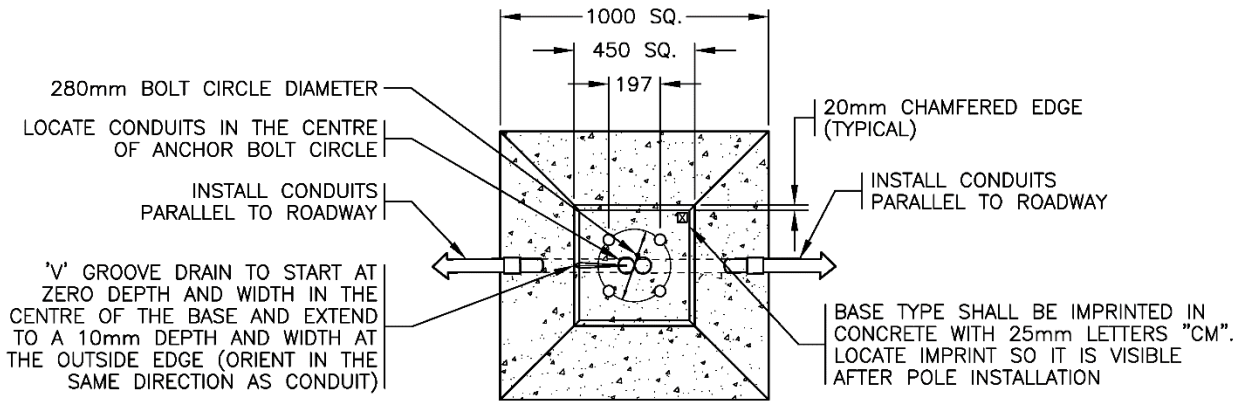
30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

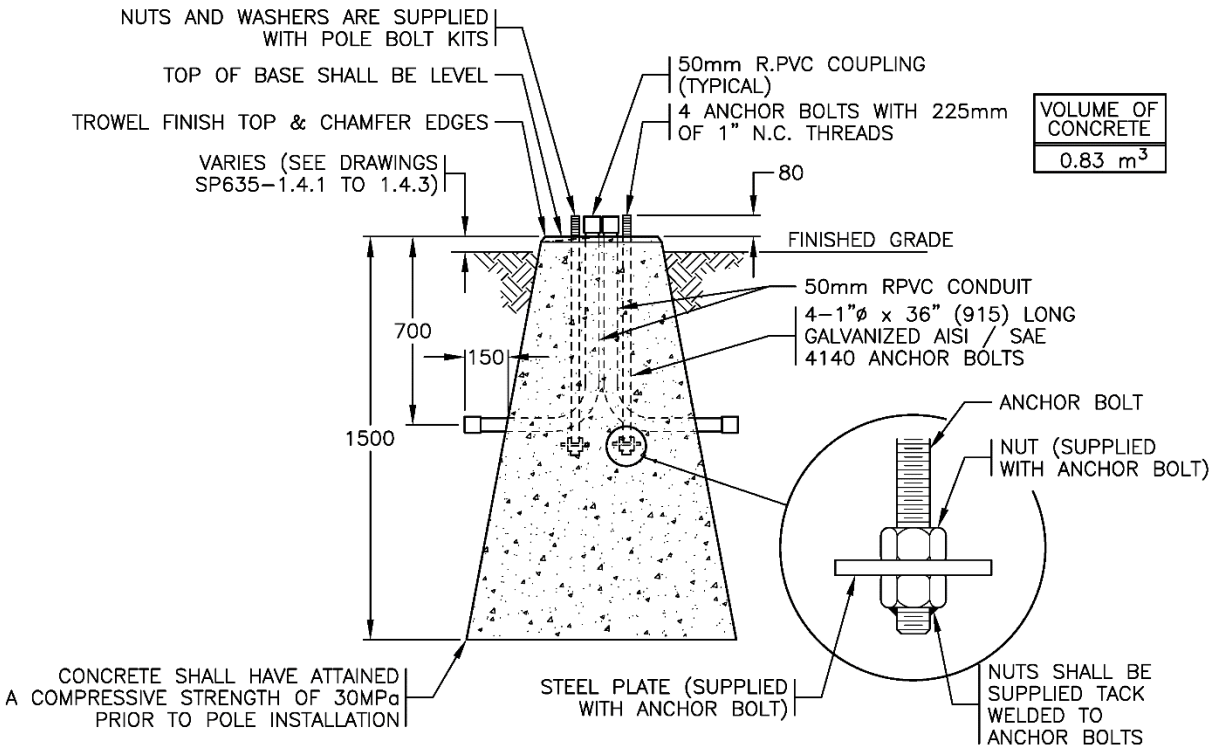
SP635-1.1.3

3



TOP VIEW

ROADSIDE



VOLUME OF CONCRETE
0.83 m^3

NOTES:

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. SEE DRAWINGS SP635-1.4.1 TO 1.4.4 FOR BACKFILL REQUIREMENTS.

BASE TYPE	POLE TYPE	APPROXIMATE MASS
CM	7.5m, 9.0m AND 11.0m TYPE 2 AND TYPES 4, 4A AND 5 POLES	2000 kg

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NOT TO SCALE

No.	Revision	Date
F		
E		
D		
C		
B	TITLE UPDATED	APR 20
A	GENERAL REVISIONS	APR 17

TYPE CM MODIFIED TYPE C CONCRETE BASE (PRECAST OR CAST IN PLACE)

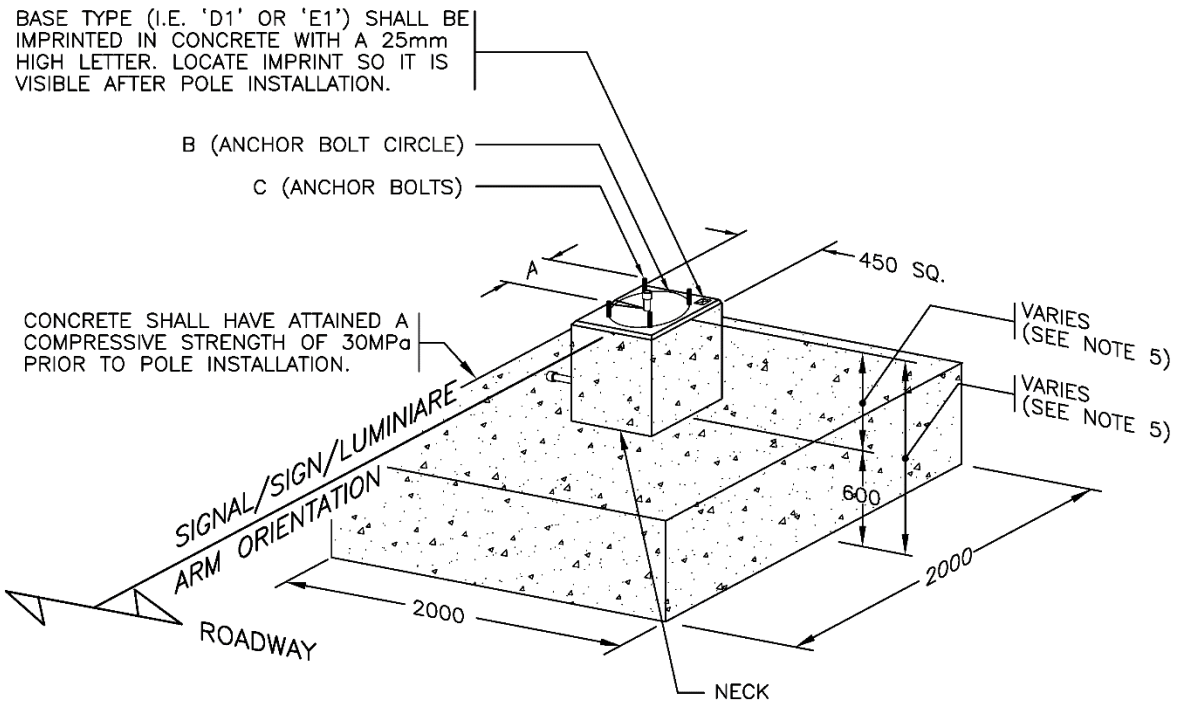
Date Approved

09/02/16 D.N. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635-1.1.3a



CONCRETE SPREAD FOOTING

BASE TYPE	POLE TYPE	VOLUME OF CONCRETE	MASS OF REBAR	FORMWORK	APPROXIMATE MASS	A	B	C (ANCHOR BOLTS)
D1	13.5m TYPE 2 POLES	2.7 m^3*	260 kg *	7.3 m^2*	6600 kg*	216	305	4–1 1/4"ø x 48" (1220) GALVANIZED AISI / SAE 4140
E1	TYPE 1 AND 3 POLES	2.7 m^3*	260 kg *	7.3 m^2*	6600 kg*	197	280	4–1"ø x 36" (915) GALVANIZED AISI / SAE 4140

(*) BASED ON SPREAD FOOTING WITH 1400 HIGH NECK

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. SEE PLANS FOR SIGNAL/SIGN/LUMINIARE ARM ORIENTATION.
3. SEE DRAWINGS SP635–1.1.5, 1.1.6 & 1.4.5 FOR ADDITIONAL DETAILS.
4. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
5. NECK HEIGHT MAY VARY FROM 750 to 1400. SEE DRAWING SP635–1.4.5 FOR MORE INFORMATION.

BASE DESIGNED FOR SOILS WITH A MINIMUM BEARING PRESSURE OF 100KPa

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D		
C		
B	TITLE UPDATED	APR 20
A	GENERAL REVISIONS	APR 17

TYPE D1 & E1 CONCRETE SPREAD FOOTINGS (PRECAST OR CAST IN PLACE)

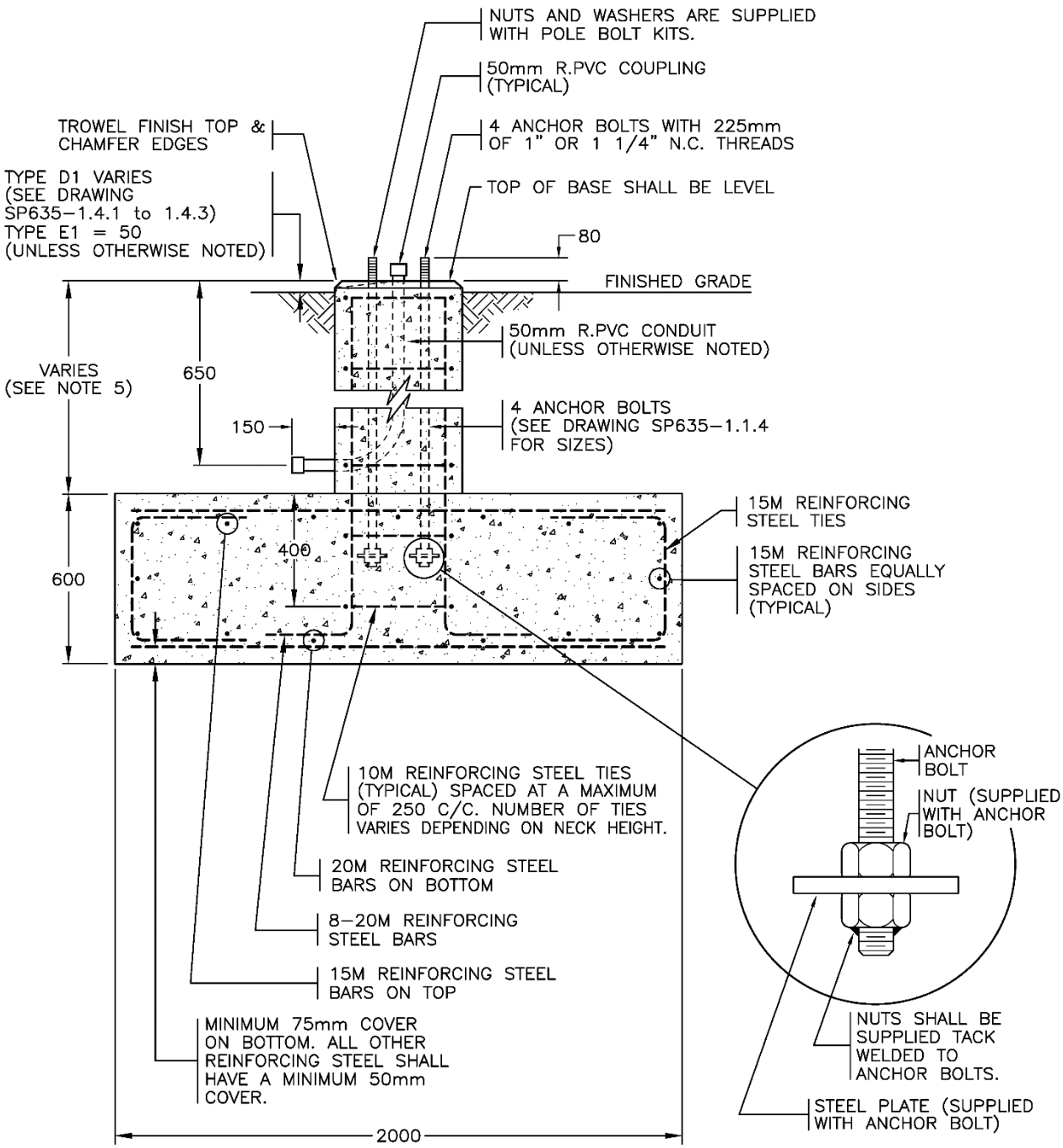
Date Approved

15/11/95 M.C. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–1.1.4



ELEVATION

SEE DRAWING SP635-1.1.4 FOR NOTES AND ADDITIONAL DETAILS

NOT TO SCALE

BRITISH COLUMBIA Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D		
C	GENERAL REVISIONS	APR 20
B	GENERAL REVISIONS	APR 17
A	ELEVATION OF BASE TO FINISHED GRADE REV'D	AUG 96

TYPE D1 & E1 CONCRETE SPREAD FOOTINGS (PRECAST OR CAST IN PLACE)

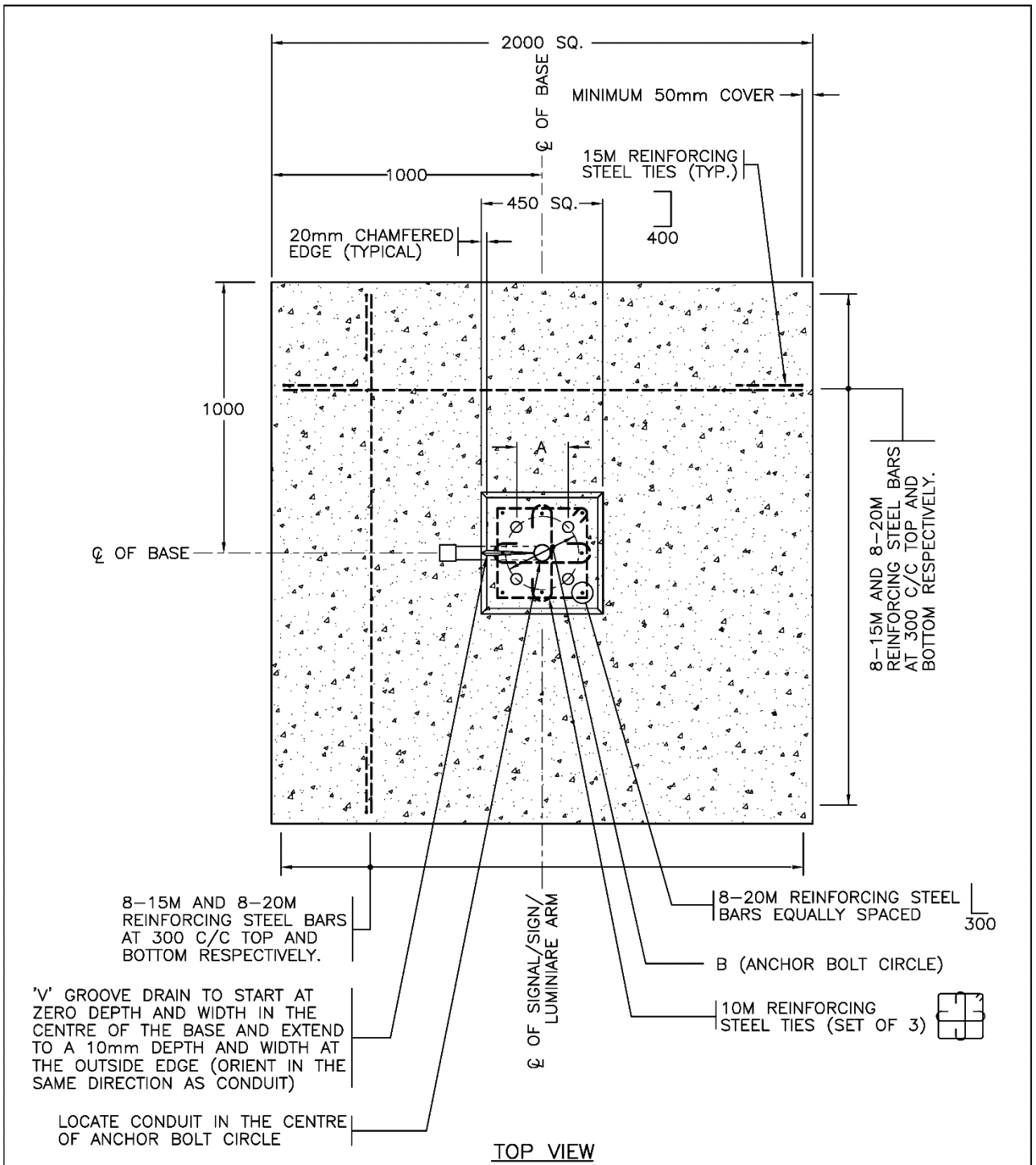
Date Approved

15/11/95 M.C. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635-1.1.5



TOP VIEW

SEE DRAWING SP635–1.1.4 FOR NOTES AND ADDITIONAL DETAILS

NOT TO SCALE

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No.	Revision	Date
F		
E		
D		
C		
B	TITLE UPDATED	APR 20
A	GENERAL REVISIONS	APR 17

TYPE D1 & E1 CONCRETE SPREAD FOOTINGS (PRECAST OR CAST IN PLACE)

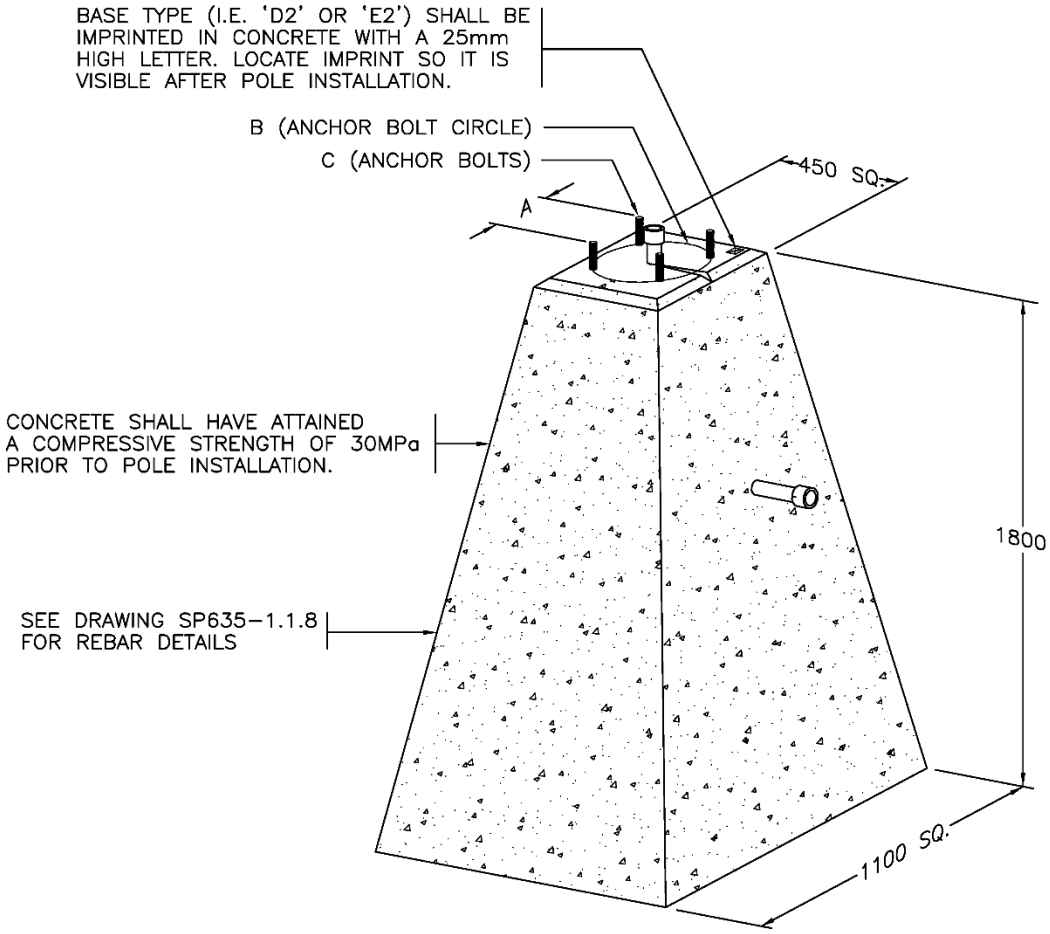
Date Approved

15/11/95 M.C. **(Signature on File)**

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–1.1.6



PRECAST CONCRETE BASES

BASE TYPE	POLE TYPE	APPROXIMATE MASS	VOLUME OF CONCRETE	A	B	C (ANCHOR BOLTS)
D2	13.5m TYPE 2 POLES	2450 kg	1.0 m^3	216	305	4–1 1/4"ø x 48" (1220) GALVANIZED AISI / SAE 4140
E2	TYPE 1 AND 3 POLES	2410 kg	1.0 m^3	197	280	4–1"ø x 36" (915) GALVANIZED AISI / SAE 4140

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. SEE DRAWING SP635–1.1.8 FOR ADDITIONAL DETAILS.
4. SEE DRAWINGS SP635–1.4.1 TO 1.4.4 FOR BACKFILL REQUIREMENTS.

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D	GENERAL REVISIONS	APR 17
C	REBAR DRAWING REFERENCE REVISED	AUG 96
B	DRAWING NUMBER CHANGED	AUG 95
A	GENERAL REVISIONS	AUG 94

TYPES D2 AND E2 CONCRETE BASES (PRECAST)

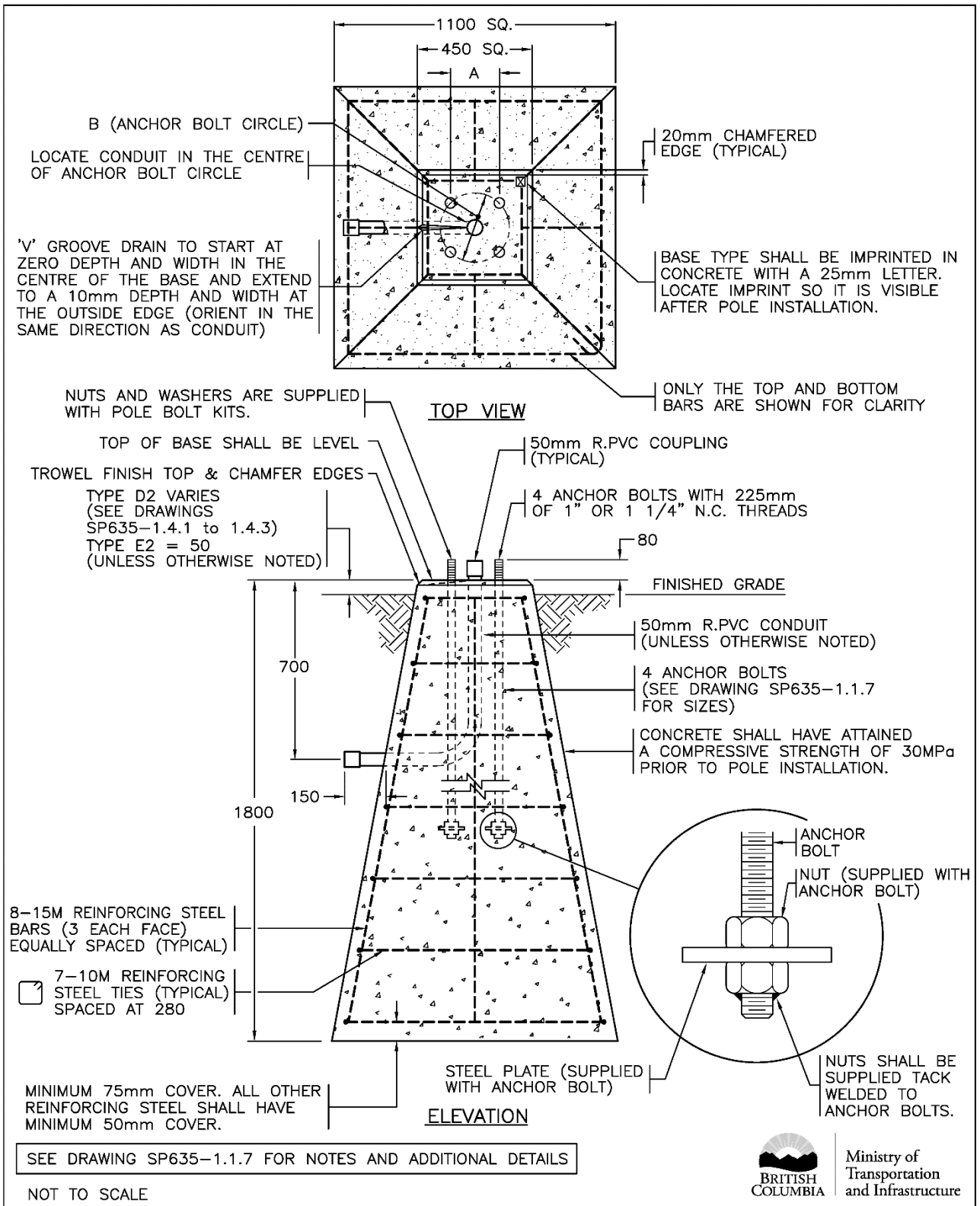
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–1.1.7



SEE DRAWING SP635–1.1.7 FOR NOTES AND ADDITIONAL DETAILS

NOT TO SCALE

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No.	Revision	Date
F		
E	GENERAL REVISIONS	APR 20
D	GENERAL REVISIONS	APR 17
C	ELEVATION OF BASE TO FINISHED GRADE REV'D	AUG 96
B	REBAR ADDED	AUG 95
A	GENERAL REVISIONS	AUG 94

TYPE D2 AND E2 CONCRETE BASES (PRECAST)

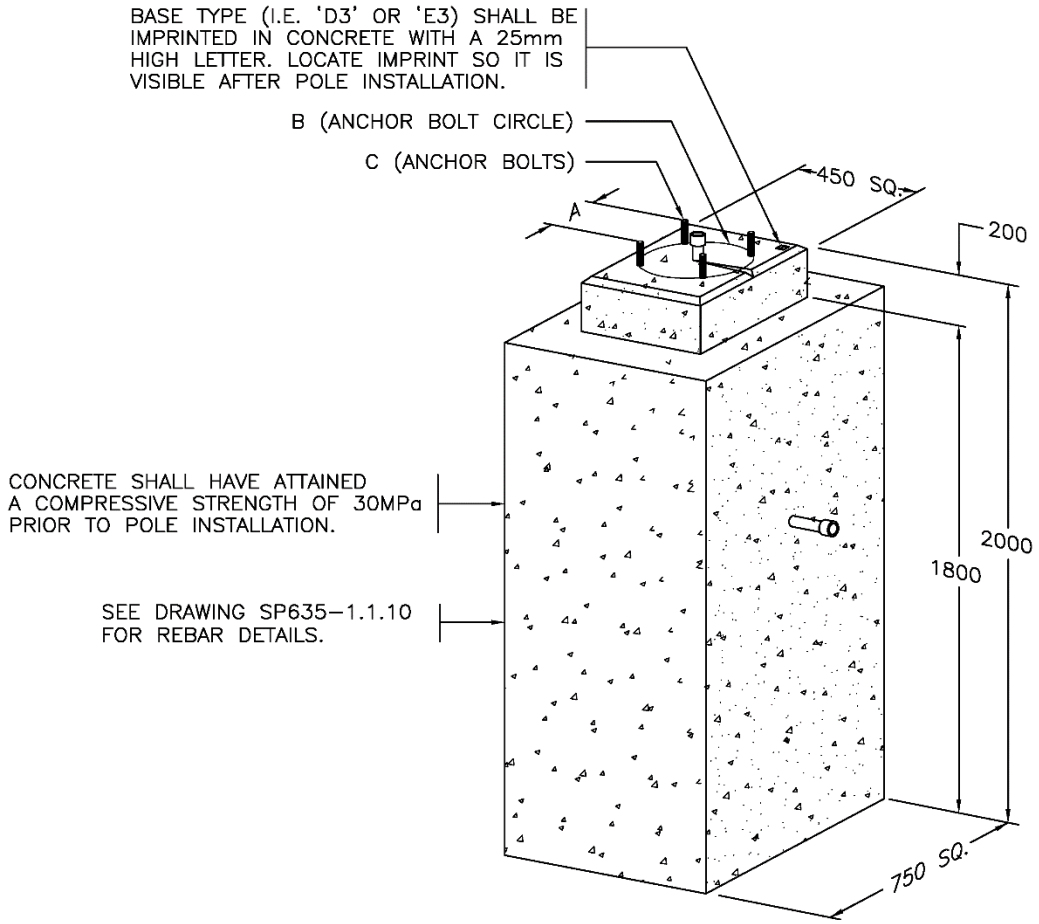
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–1.1.8



POURED IN PLACE CONCRETE BASES

BASE TYPE	POLE TYPE	APPROXIMATE MASS	VOLUME OF CONCRETE	A	B	C (ANCHOR BOLTS)
D3	13.5m TYPE 2 POLES	2550 kg	1.05 m^3	216	305	4–1 1/4"ø x 48" (1220) GALVANIZED AISI / SAE 4140
E3	TYPE 1 AND 3 POLES	2510 kg	1.05 m^3	197	280	4–1"ø x 36" (915) GALVANIZED AISI / SAE 4140

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. SEE DRAWINGS SP635–1.1.10 FOR ADDITIONAL DETAILS.
4. SEE DRAWINGS SP635–1.4.1 TO 1.4.4 FOR BACKFILL REQUIREMENTS.

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D	TITLE UPDATED	APR 20
C	GENERAL REVISIONS	APR 17
B	DRAWING NUMBER CHANGED	AUG 95
A	GENERAL REVISIONS	AUG 94

TYPE D3 AND E3 CONCRETE BASES (CAST IN PLACE)

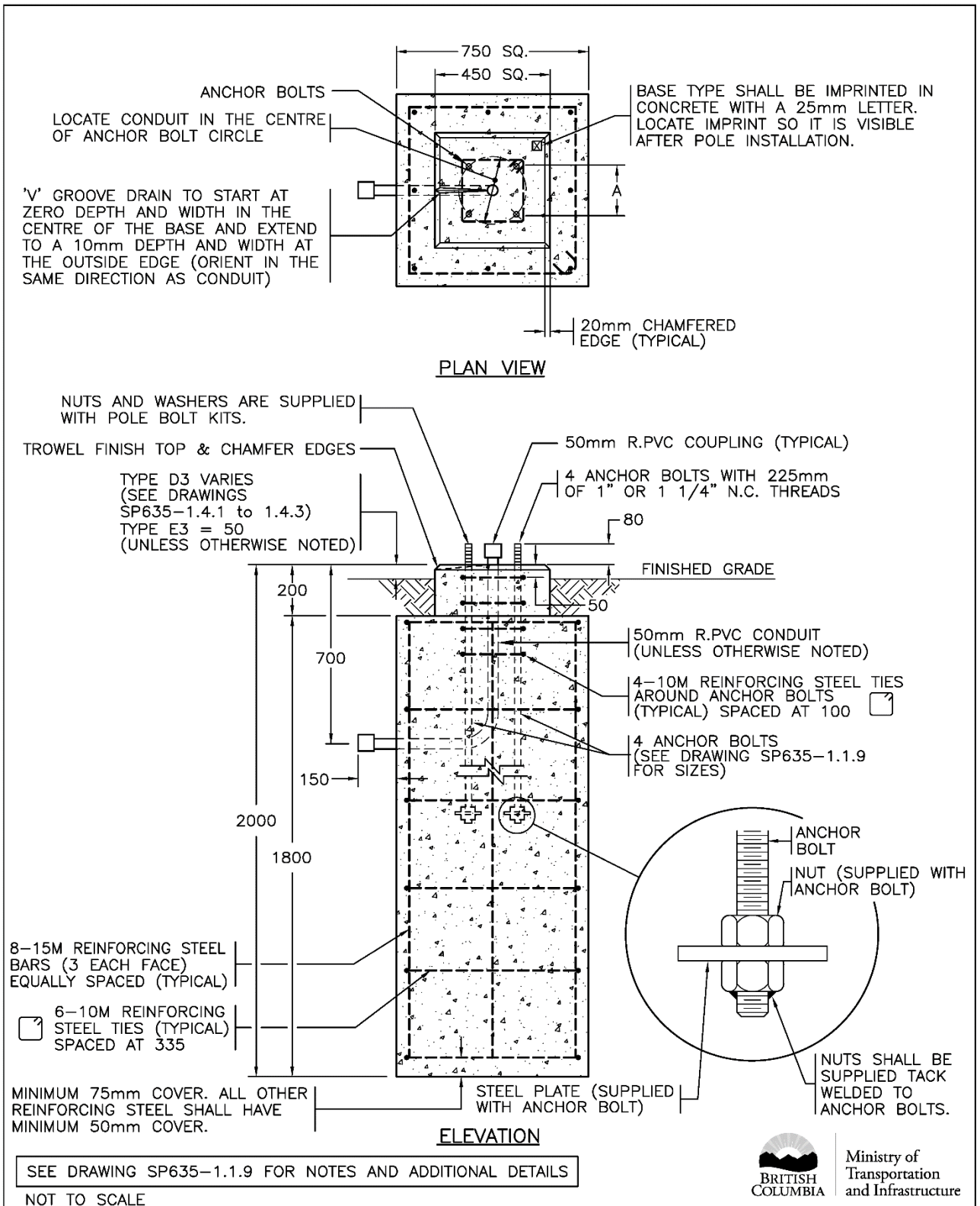
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–1.1.9



SEE DRAWING SP635-1.1.9 FOR NOTES AND ADDITIONAL DETAILS

NOT TO SCALE

BRITISH COLUMBIA Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E	TITLE UPDATE	APR 20
D	GENERAL REVISIONS	APR 17
C	ELEVATION OF BASE TO FINISHED GRADE REV'D	AUG 96
B	REBAR ADDED	AUG 95
A	GENERAL REVISIONS	AUG 94

TYPE D3 AND E3 CONCRETE BASES (CAST IN PLACE)

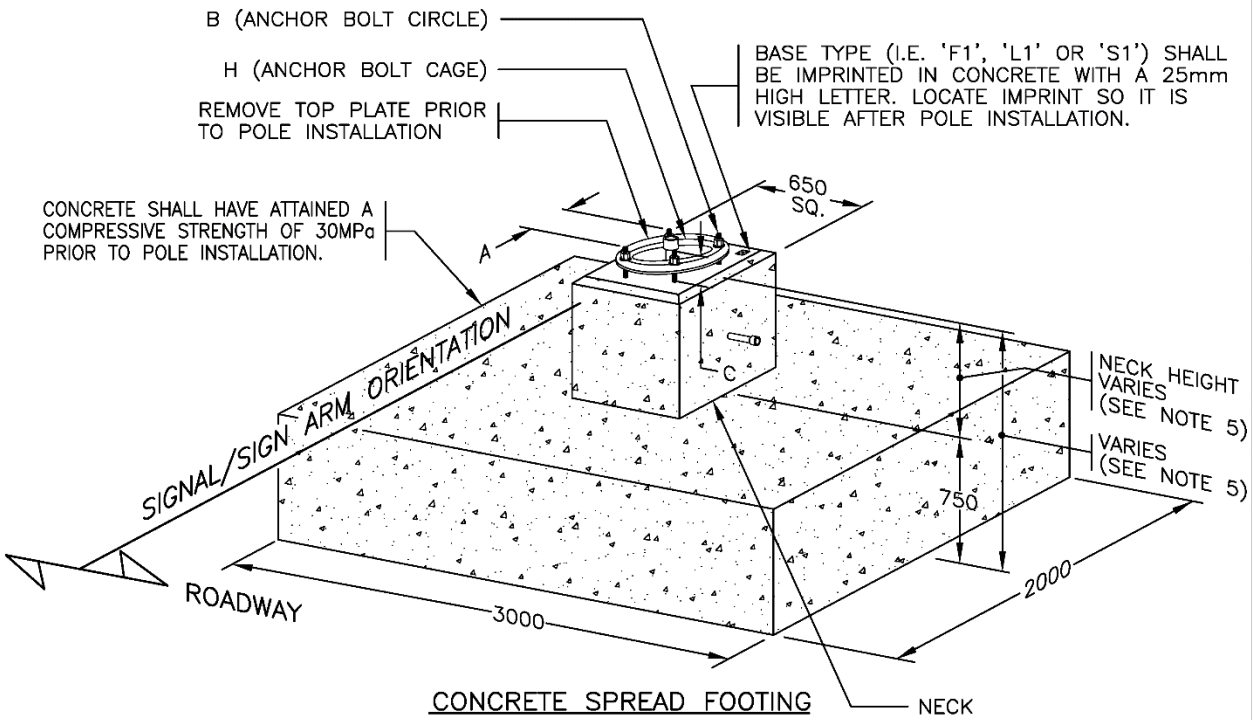
Date Approved

30/09/93 E.L. **(Signature on File)**

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635-1.1.10



CONCRETE SPREAD FOOTING

BASE TYPE	POLE TYPE	VOLUME OF CONCRETE	MASS OF REBAR	APPROXIMATE MASS	FORMWORK	A	B	C	H (ANCHOR BOLTS)
F1	TYPE 6 AND 7 POLES	5.0m^3	455 kg*	12306 kg*	11.0m^2	243	343	160	4–1"ø x 48" (1220) GALVANIZED GRADE 150 DYWIDAG PRE–ASSEMBLED IN A CAGE
S1	TYPE S POLES	5.0m^3	455 kg*	12306 kg*	11.0m^2	243	343	160	4–1"ø x 48" (1220) GALVANIZED GRADE 150 DYWIDAG PRE–ASSEMBLED IN A CAGE
L1	TYPE L POLES	5.0m^3	455 kg*	12306 kg*	11.0m^2	276	390	140	4–1 1/2"ø x 54" (1370) GALVANIZED AISI / SAE 4140 PRE–ASSEMBLED IN A CAGE

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. SEE PLANS FOR SIGNAL/SIGN ARM ORIENTATION.
3. SEE DRAWINGS SP635–1.1.12, 1.1.13 & 1.4.5 FOR ADDITIONAL DETAILS.
4. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
5. NECK HEIGHT MAY VARY FROM 750 to 1400. SEE DRAWING SP635–1.4.5 FOR MORE INFORMATION.

BASE DESIGNED FOR SOILS WITH A MINIMUM BEARING PRESSURE OF 100KPa

NOT TO SCALE

BRITISH COLUMBIA Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D		
C	TITLE UPDATE	APR 20
B	GENERAL REVISIONS	APR 17
A	TYPE S1 BASE ADDED	OCT 03

TYPE F1, L1 & S1 CONCRETE SPREAD FOOTINGS (CAST IN PLACE)

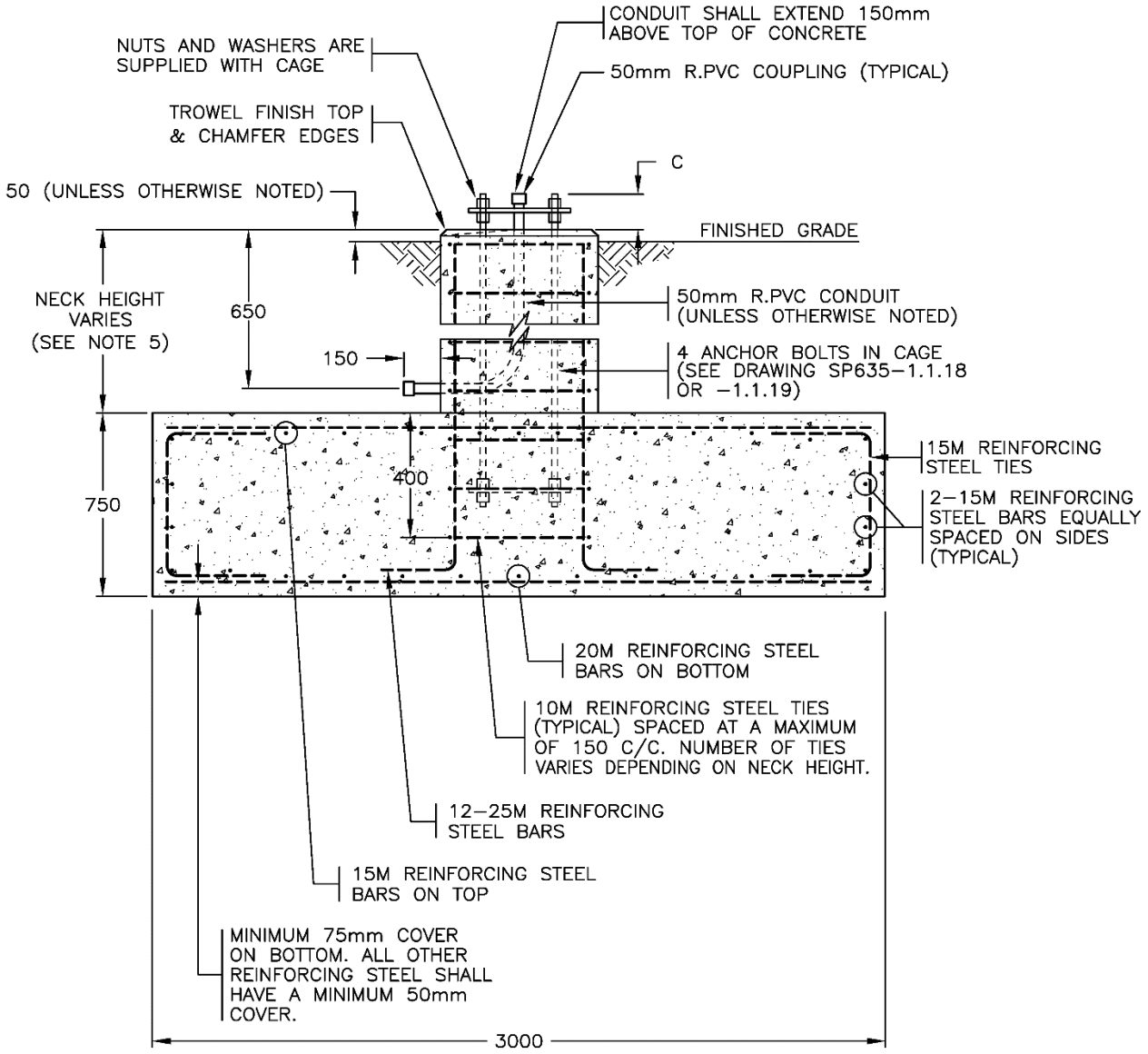
Date Approved

15/11/95 M.C. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–1.1.11



ELEVATION

SEE DRAWING SP635-1.1.11 FOR NOTES AND ADDITIONAL DETAILS

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D	TITLE UPDATED	APR 20
C	GENERAL REVISIONS	APR 17
B	TYPE S1 BASE ADDED	OCT 03
A	ELEVATION OF BASE TO FINISHED GRADE REV'D	AUG 96

TYPE F1, L1 & S1 CONCRETE SPREAD FOOTINGS (CAST IN PLACE)

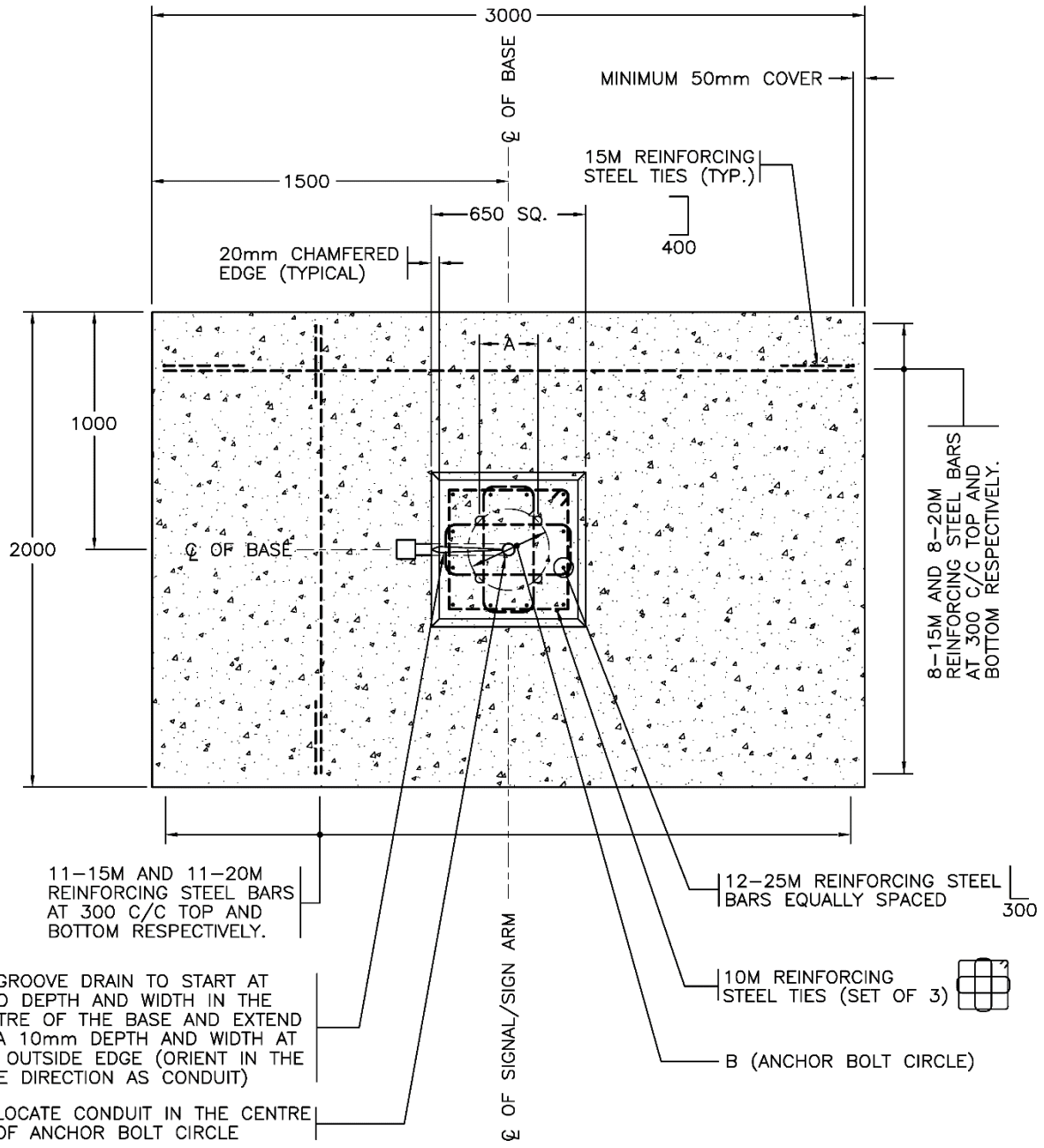
Date Approved

15/11/95 M.C. **(Signature on File)**

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635-1.1.12



TOP VIEW

SEE DRAWING SP635-1.1.11 FOR NOTES AND ADDITIONAL DETAILS

NOT TO SCALE

Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D		
C	TITLE UPDATED	APR 20
B	GENERAL REVISIONS	APR 17
A	TYPE S1 BASE ADDED	OCT 03

TYPE F1, L1 & S1 CONCRETE SPREAD FOOTINGS (CAST IN PLACE)

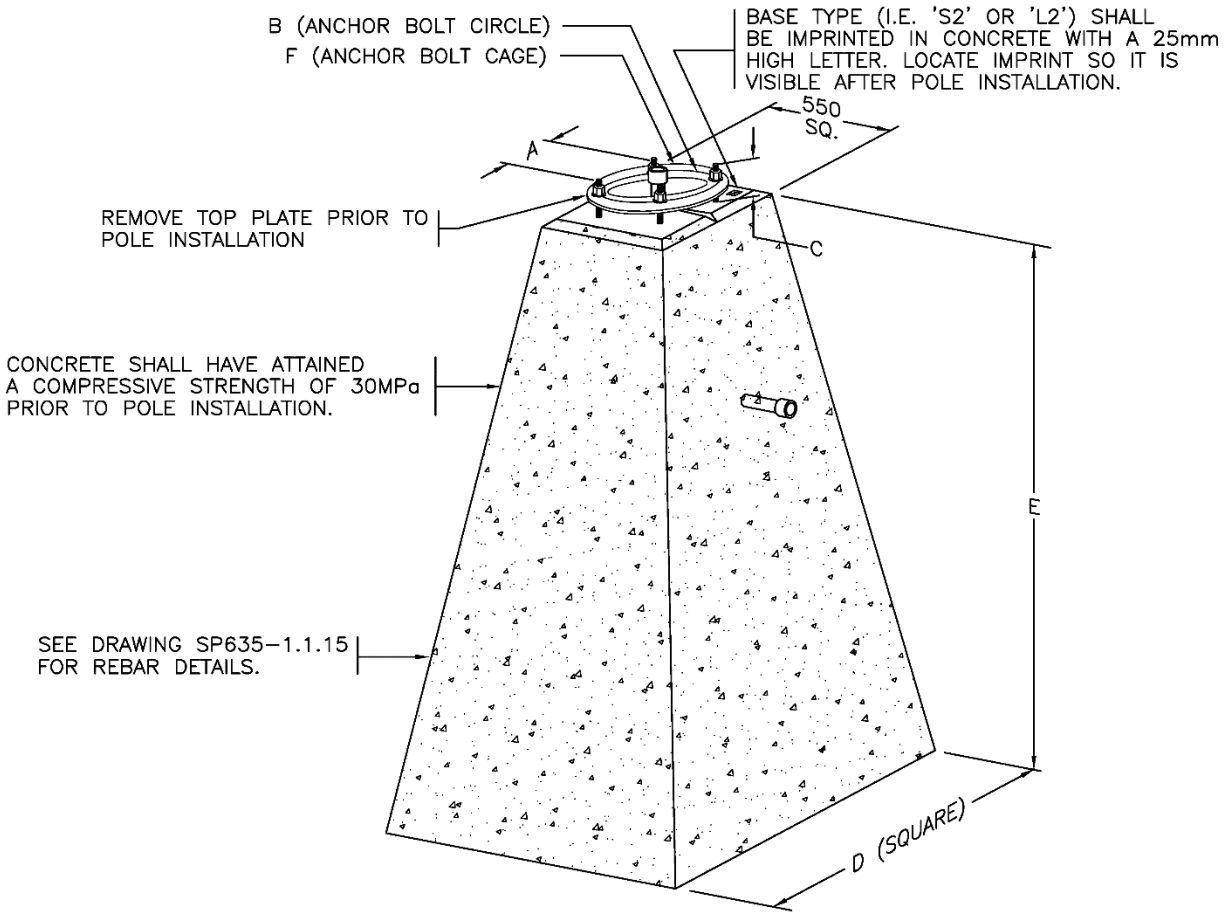
Date Approved

15/11/95 M.C. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635-1.1.13



PRECAST CONCRETE BASES

BASE TYPE	POLE TYPE	APPROXIMATE MASS	VOLUME OF CONCRETE	A	B	C	D	E	F (ANCHOR BOLTS)
S2 F2	TYPE S, 6 AND 7 POLES	4500 kg	1.8 m^3	243	343	160	1240	2100	4–1"ø x 48" (1220) GALVANIZED GRADE 150 DYWIDAG PRE–ASSEMBLED IN A CAGE
L2	TYPE L POLES	5040 kg	2.0 m^3	276	390	140	1300	2300	4–1 1/2"ø x 54" (1370) GALVANIZED AISI/SAE 4140 PRE–ASSEMBLED IN A CAGE

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. SEE DRAWING SP635–1.1.15 FOR ADDITIONAL DETAILS.
4. SEE DRAWINGS SP635–1.4.1 TO 1.4.4 FOR BACKFILL REQUIREMENTS.

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E	COMBINED S2 AND F2 BASE	APR 20
D	GENERAL REVISIONS	APR 17
C	TYPE S2 BASE ADDED	OCT 03
B	DRAWING NUMBER CHANGED	AUG 95
A	GENERAL REVISIONS	AUG 94

TYPE F2, L2 AND S2 CONCRETE BASES (PRECAST)

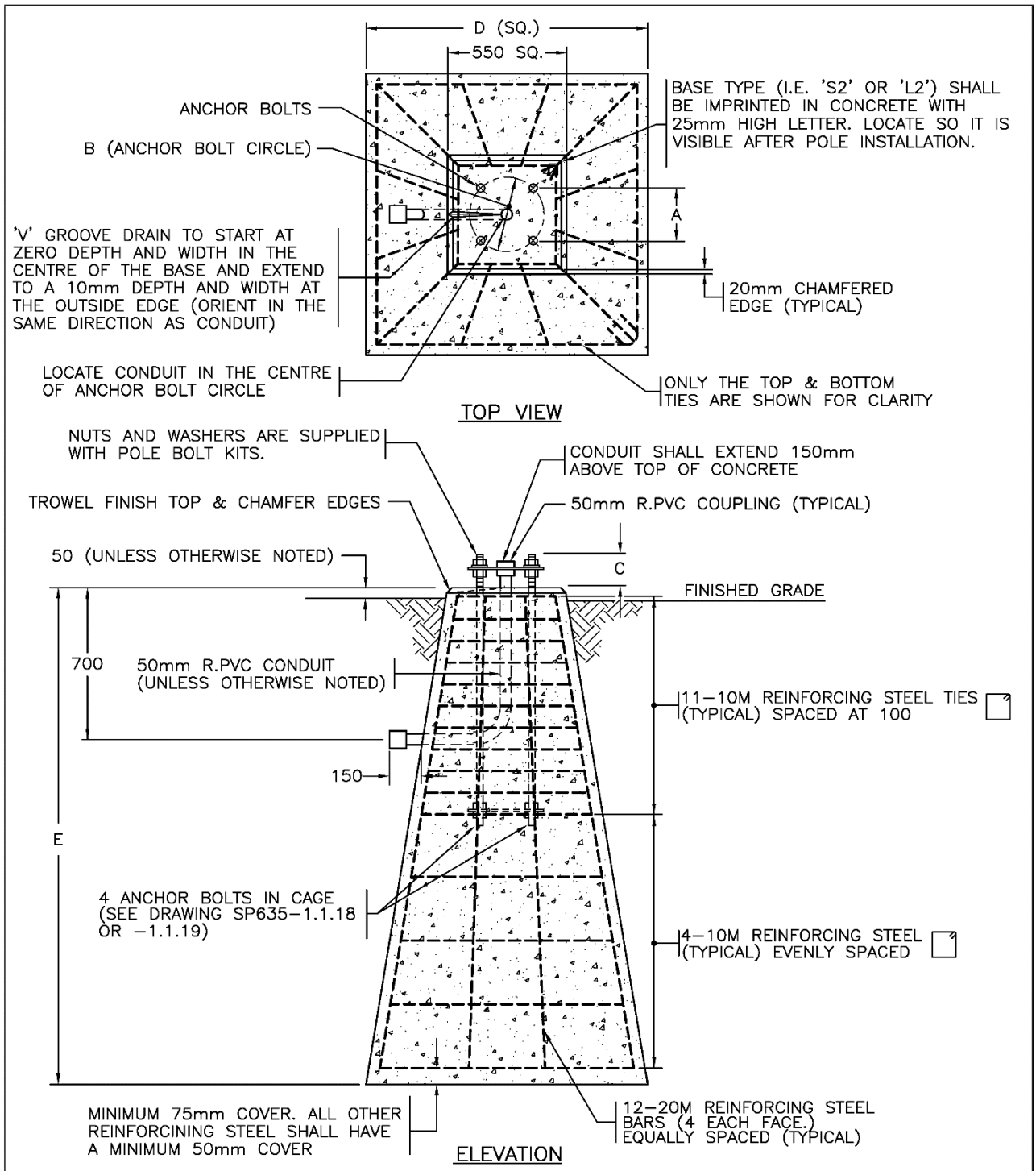
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–1.1.14



SEE DRAWING SP635–1.1.14 FOR NOTES AND ADDITIONAL DETAILS

NOT TO SCALE

BRITISH COLUMBIA Ministry of Transportation and Infrastructure

No.	Revision	Date
F	COMBINED S2 AND F2 BASE	APR 20
E	GENERAL REVISIONS	APR 17
D	TYPE S2 BASE ADDED	OCT 03
C	ELEVATION OF BASE TO FINISHED GRADE REVISED	AUG 96
B	ELEVATION OF BASE TO FINISHED GRADE REVISED	AUG 95
A	GENERAL REVISIONS	AUG 94

TYPE F2, L2 AND S2 CONCRETE BASES (PRECAST)

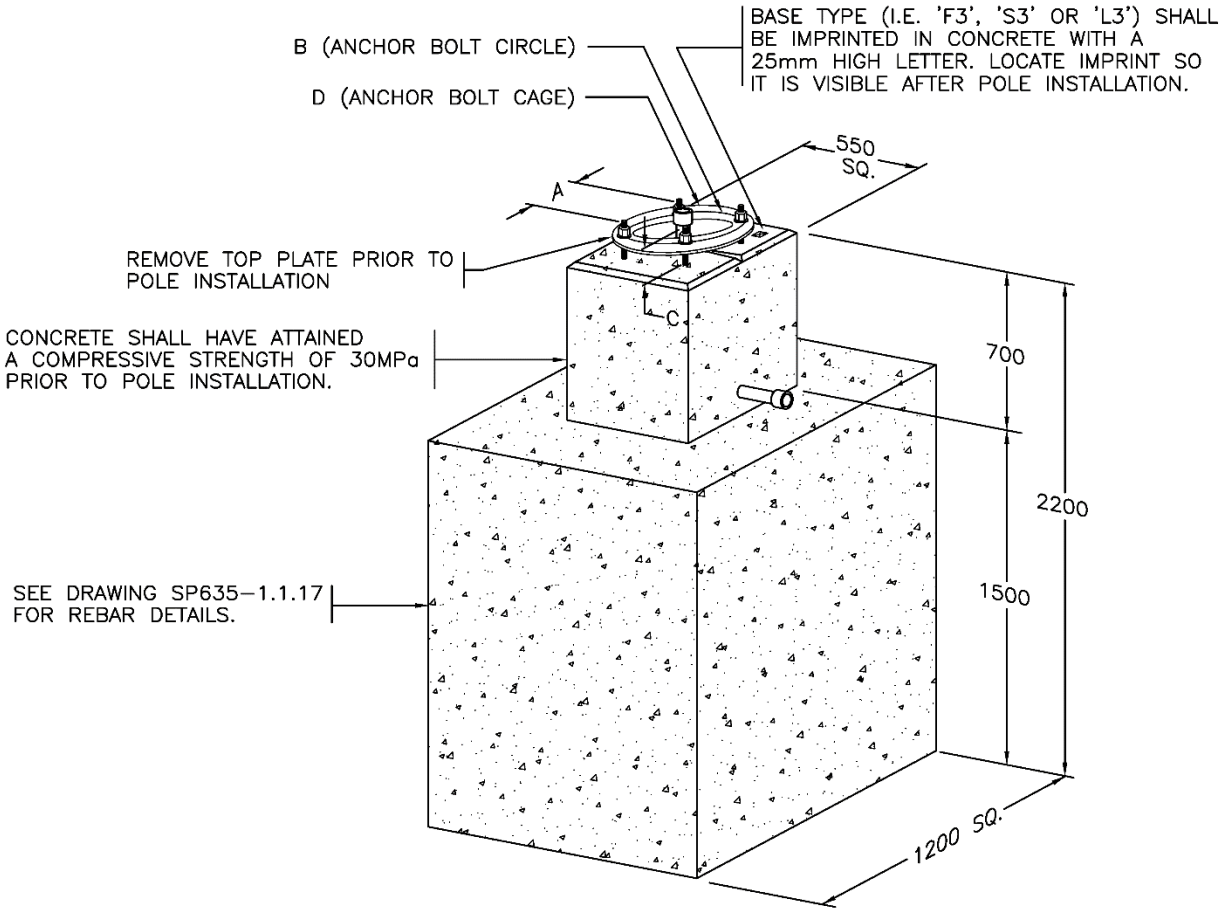
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–1.1.15



POURED IN PLACE CONCRETE BASES

BASE TYPE	POLE TYPE	APPROXIMATE MASS	VOLUME OF CONCRETE	A	B	C	D (ANCHOR BOLTS)
F3	TYPE 6 AND 7 POLES	5925 kg	2.37m^3	243	343	160	4–1"ø x 48" (1220) GALVANIZED GRADE 150 DYWIDAG BOLTS PRE–ASSEMBLED IN A CAGE
S3	TYPE S POLES	5925 kg	2.37m^3	243	343	160	4–1"ø x 48" (1220) GALVANIZED GRADE 150 DYWIDAG BOLTS PRE–ASSEMBLED IN A CAGE
L3	TYPE L POLES	5965 kg	2.37m^3	276	390	140	4–1 1/2"ø x 54" (1370) GALVANIZED AISI / SAE 4140 BOLTS PRE–ASSEMBLED IN A CAGE

NOTES

1. SEE STANDARD SPECIFICATIONS & SPECIAL PROVISIONS FOR ADDITIONAL INFORMATION.
2. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED.
3. SEE DRAWING SP635–1.1.17 FOR ADDITIONAL DETAILS.
4. SEE DRAWINGS SP635–1.4.1 TO 1.4.4 FOR BACKFILL REQUIREMENTS.

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NOT TO SCALE

No.	Revision	Date
F		
E	GENERAL REVISIONS	APR 20
D	GENERAL REVISIONS	APR 17
C	TYPE S3 BASE ADDED	OCT 03
B	DRAWING NUMBER CHANGED	AUG 95
A	GENERAL REVISIONS	AUG 94

TYPE F3, L3 & S3 CONCRETE BASES (CAST IN PLACE)

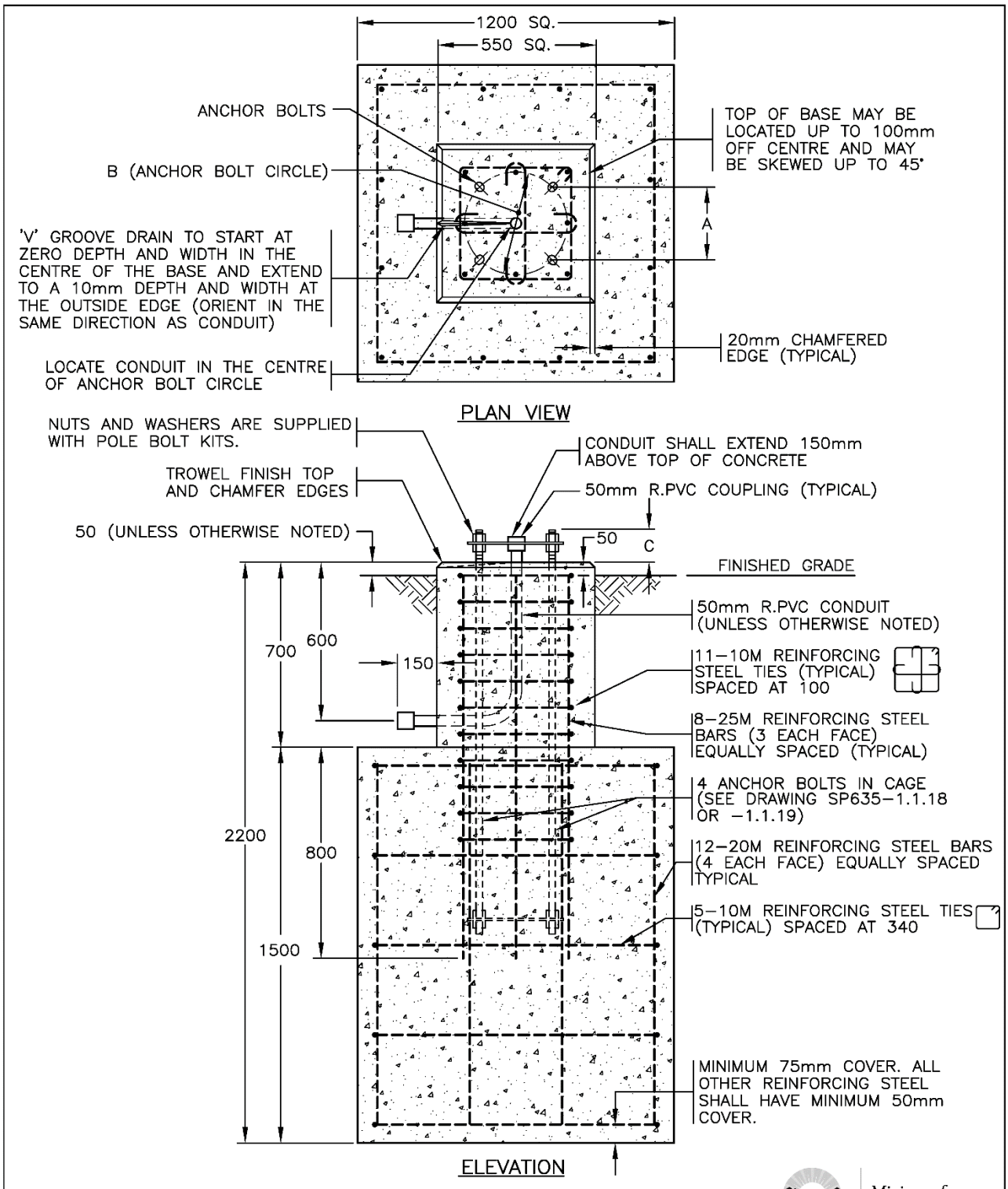
Date Approved

30/09/93 E.L. (Signature on File)

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–1.1.16



SEE DRAWING SP635–1.1.16 FOR NOTES AND ADDITIONAL DETAILS

NOT TO SCALE

BRITISH COLUMBIA — Ministry of Transportation and Infrastructure

No.	Revision	Date
F	TITLE UPDATED	APR 20
E	GENERAL REVISIONS	APR 17
D	TYPE S3 BASE ADDED	OCT 03
C	ELEVATION OF BASE TO FINISHED GRADE REVISED	AUG 96
B	NECK WIDTH REVISED & REBAR ADDED	AUG 95
A	GENERAL REVISIONS	AUG 94

TYPE F3, L3 & S3 CONCRETE BASES (CAST IN PLACE)

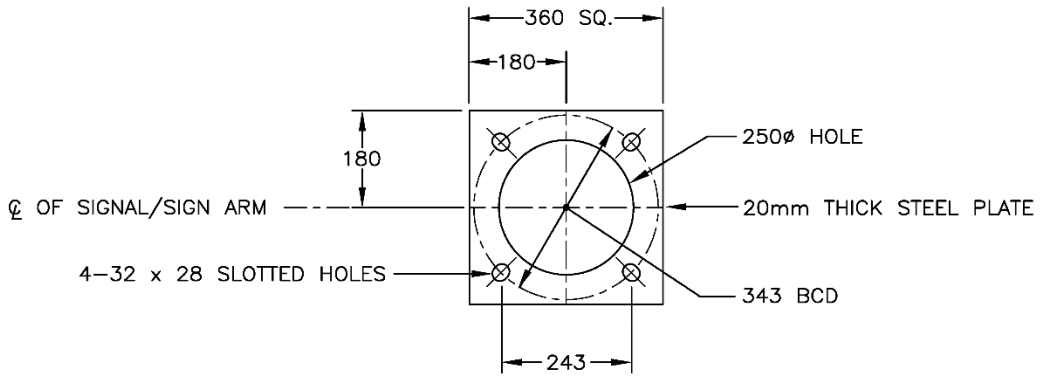
Date Approved

30/09/93 E.L. (Signature on File)

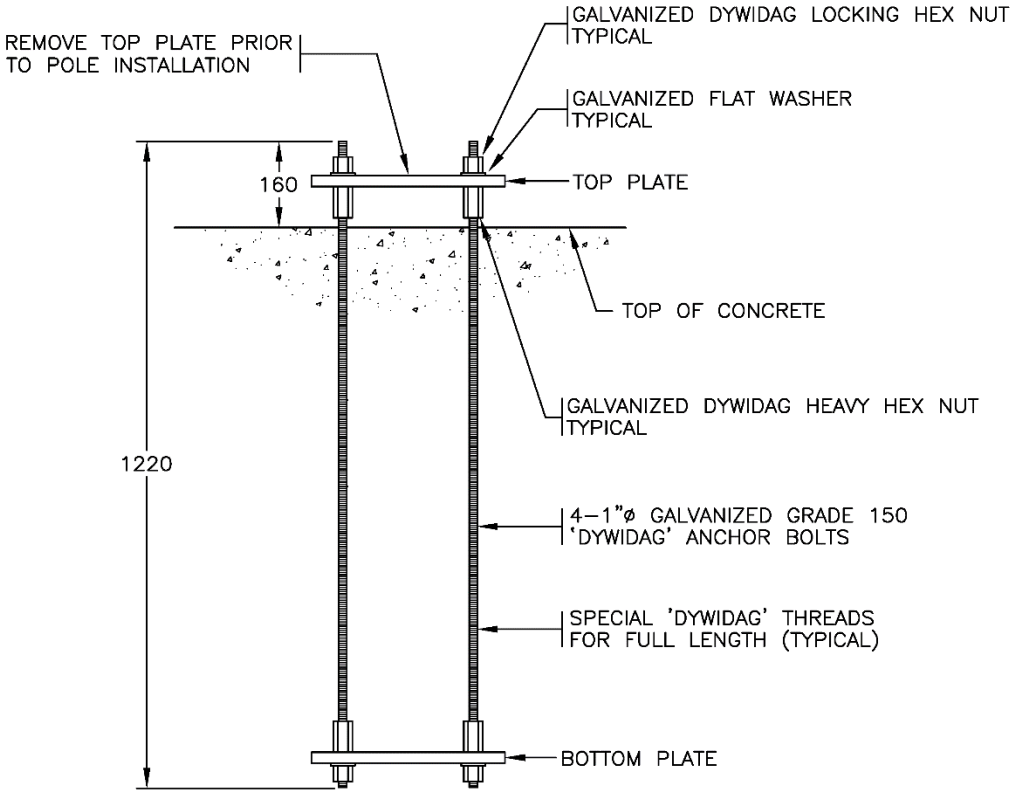
Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–1.1.17



TOP AND BOTTOM PLATES



ANCHOR BOLT CAGE

APPROXIMATE MASS
60 kg

SEE DRAWINGS SP635–1.1.11, 1.1.14 AND 1.1.16 FOR NOTES AND ADDITIONAL DETAILS

NOT TO SCALE

BRITISH COLUMBIA | Ministry of Transportation and Infrastructure

No.	Revision	Date
F		
E		
D	GENERAL REVISIONS	APR 17
C	TYPE S POLE ADDED & TOP PLATE NOTE REVISED	OCT 03
B	HOLE SIZE IN PLATE ADDED	AUG 95
A	GENERAL REVISIONS	AUG 94

ANCHOR BOLT CAGE FOR TYPE 6, 7 & S POLES

Date Approved

30/09/93 E.L. **(Signature on File)**

Chief Highway Engineer

SPECIFICATION DRAWING No.

SP635–1.1.18